PREP 1, LESSONS 1–2

Complete circuits and current

Circuit symbols; ammeters; current as a flow of charge; Q = I t

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the law of conservation of energy. DN 2 [1]

Energy can be transferred usefully, stored or dissipated, but it cannot be created or destroyed.

2 Define power. DN 31 [1]

The rate at which energy is transferred (or work is done).

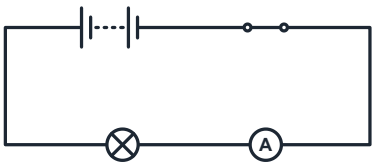
3 Describe how a line of best fit should be drawn. DN 893 [1]

One thin straight line or smooth curve, close to the points, with about as many above it as below.

Key ideas LESSONS 1–2

Show your working

4 Look at this circuit.



a The ammeter is connected in (series) / parallel with the lamp. [1]

b The switch is opened. Explain why the lamp goes out. [1]

The circuit is no longer a complete loop, so there is no current ✓

c Describe the current at different points in this loop. [1]

It is the same at every point ✓

5 A current of 0.40 A flows through a lamp for 60 s. Use Q = I t

Calculate the charge that flows through the lamp. [3]

Q = I t ✓

Q = 0.40 × 60 ✓

Q = 24 C ✓

6 State what is meant by electric current. [1]

The rate of flow of (electrical) charge ✓